Lecture Notes on

Composition of the Universe

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These notes summarize the tenth Astrophysics Club meeting, where we explored the composition of the universe: ordinary matter, dark matter, and dark energy. The central theme is that 95% of the universe remains something we do not fully understand.

Contents

1 Cosmology

Cosmology is the study of the origin, evolution, and composition of the universe as a whole.

Essential question:

- What is the universe made out of?

For centuries, humans have tried to answer this — from ancient ideas of earth, water, air, and fire to the Standard Model of particle physics.

Modern physics tells us the universe has three main components:

- Ordinary (baryonic) matter
- Dark matter
- Dark energy

The striking fact: the majority of the universe is not visible.

2 The Composition Breakdown

- Ordinary matter: $\sim 5\%$
- Dark matter: $\sim 27\%$
- Dark energy: $\sim 68\%$

This means roughly 95% of the universe is something we do not fully understand.

3 Ordinary (Baryonic) Matter

Everything we can directly observe is made of atoms — protons, neutrons, and electrons.

Forms:

- Stars
- Planets
- Gas and dust

Key facts:

- Only $\sim 5\%$ of the universe.
- The *only* type of matter that interacts with light.

4 Dark Matter

We cannot see dark matter, but we can see its effects.

Properties:

- Does not emit or absorb light.
- Does not interact electromagnetically.
- Only interacts through gravity.
- Comprises $\sim 27\%$ of the universe — about $5\times$ more than ordinary matter.

5 Evidence for Dark Matter

5.1 Galaxy Rotation Curves

The expected orbital velocity of a star at distance r from the center of a galaxy, based on visible mass alone, is

$$v(r) = \sqrt{\frac{GM(r)}{r}},$$

where $G = 6.674 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ is the universal gravitational constant and $M(r)$ is the mass enclosed within radius r.

The problem:

- Stars at the outer edges of galaxies orbit much faster than predicted by visible mass alone.
- Expected: $v(r)$ decreases with distance (like planets in our Solar System).
- Observed: $v(r)$ stays roughly constant at large r.

Conclusion: there must be unseen mass. This was one of the first major clues for dark matter.

5.2 Gravitational Lensing

Massive objects bend the path of light passing near them. This is predicted by general relativity.

Einstein's deflection angle (1936):

$$\theta = \frac{4GM}{bc^2},$$

where M is the mass of the lensing object, b is the impact parameter (closest approach distance), and c is the speed of light.

What this means:

- We observe light bending around regions of invisible mass.
- The amount of bending tells us how much dark matter is present.
- Predicted by general relativity — confirmed by observation.

6 Why Dark Matter Matters

Dark matter acts like an invisible framework holding the universe together.

- Provides extra gravitational mass to galaxies.
- Prevents galaxies from flying apart.
- Forms large-scale structure (the cosmic web).

Without dark matter, galaxies as we know them could not exist.

7 Dark Energy

Dark energy is the most abundant and least understood component of the cosmos.

Properties:

- Drives the accelerated expansion of the universe.
- Works opposite to gravity on large scales.
- Discovered in 1998 using distant Type Ia supernovae as "standard candles."
- Comprises $\sim 68\%$ of the universe.

8 The Friedmann Equation

The Friedmann equation governs how the universe expands — it is the master equation of cosmology:

$$H^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}.$$

Where:

- H — Hubble parameter (expansion rate).
- ρ — energy density (matter + radiation).
- k — spatial curvature.
- Λ — cosmological constant (dark energy).
- a — scale factor of the universe.

All three components — matter, curvature, and dark energy — compete to determine the fate of the universe.

Connection to earlier meetings: in Meeting 2 we introduced Hubble's Law, $v = H_0 d$. The Friedmann equation is the deeper result from which Hubble's Law emerges.

9 Density Parameters

The Friedmann equation simplifies into a balance of dimensionless density parameters:

$$\Omega_m + \Omega_\Lambda + \Omega_k = 1.$$

- Ω_m — matter density (ordinary + dark matter).
- Ω_Λ — dark energy density.
- Ω_k — curvature contribution.

Measured values (Planck 2018):

Ω_m	≈ 0.315	Matter (ordinary + dark)
Ω_Λ	≈ 0.685	Dark energy
Ω_k	≈ 0	Curvature (flat universe!)

The near-zero curvature tells us the universe is spatially flat to high precision.

10 Bonus: Exotic Matter

Under extreme conditions (very high pressure, density, or temperature), matter can take exotic forms.

Found in:

- Neutron stars.
- The early universe (right after the Big Bang).

Examples:

- **Neutron-degenerate matter** — protons and electrons are crushed together into neutrons under immense gravitational pressure.
- **Quark matter** — at even higher densities, particles break down into free quarks.

11 Bonus: Strange Matter

Strange matter is a type of quark matter composed of:

- Up quarks
- Down quarks
- Strange quarks (heavier, normally unstable)

Normally, quarks are locked inside protons and neutrons. Under extreme pressure, they may break free and form a "quark soup." Adding strange quarks might make the matter *more* stable — but this remains theoretical.

12 What We Still Don't Know

- What is dark matter made of?
- What is dark energy?
- Is the cosmological constant truly constant?
- Does strange matter exist?

These are among the biggest open questions in physics today.

13 Big Picture

This meeting connects to our earlier work:

- Meeting 2 introduced Hubble's Law and the expanding universe.
- Meeting 8 covered black holes, where extreme gravity dominates.
- Now we see the full cosmic inventory: what the universe is made of and how it evolves.

The central takeaway: 95% of the universe is a mystery. Understanding dark matter and dark energy is one of the greatest challenges in modern physics.